ISE 218: Fundamentals of Information Technology

Stony Brook Institute at Anhui University

Homework #4

Spring 2025

Assignment Due: April 11, 2025

Neatly write or type your answers to the following problems on a separate sheet of paper and submit a digital copy on Microsoft Teams by the due date. For mathematical calculations you must show all work to receive full credit and simplify your answers as much as possible, giving a single number as your answer.

Please clearly print or type your full name and ID # on the first page of your submission.

For all questions involving sizes, you should use 2-based units of size. For example, 1 KB = 2^{10} bytes, not 10^3 bytes.

1. [5 pts] Show how the value 0x29AB01 would be stored by byte-addressable machines with a 32-bit integer, using little-endian and then big-endian format. Assume the value starts at address 00

   Little-endian:
   Address 00: 0x01
   Address 01: 0xAB
   Address 02: 0x29
   Address 03: 0x00 (padding to 32 bits)
   Big-endian:
   Address 00: 0x00 (padding to 32 bits)
   Address 01: 0x29
   Address 02: 0xAB
   Address 03: 0x01

2. [5 pts] What is the instruction pipeline and explain the reasons for instruction pipeline may stall?

   An **instruction pipeline** is a technique where multiple instruction phases (fetch, decode, execute, etc.) are overlapped to improve performance.
   Reasons for pipeline stalls:
   Data hazards (when instructions depend on each other's results),
   Control hazards (caused by branches),
   Structural hazards (when hardware resources are insufficient).

3. [5 pts] A computer has a memory unit with 16 bits per word. The instruction set consists of 135 different operations and has 32 general-purpose registers. Instructions have an opcode part, and two address parts. Each instruction is stored in one word of memory. How many bits are needed to specify a memory address?

Opcode: $\lceil \log_2(135) \rceil = 8$ bits
Register: $\lceil \log_2(32) \rceil = 5$ bits
Total word size = 16 bits
Let x = number of bits for each address.
Instruction format: 8 (opcode) + x (address1) + x (address2) $\leqslant$ 16
$\rightarrow 8 + 2x \leqslant 16$
$\rightarrow 2x \leqslant 8$
$\rightarrow x \leqslant 4$
Answer: 4 bits are needed for a memory address.

4. [6 pts] For the instruction LOAD 700, what value is loaded into the accumulator for each addressing mode?
Memory:

Address	**Contents**
0x700	0x800
...	
0x800	0x200
...	
0x1000	0x500
...	
0x1100	0x700
...	
0x1200	0x800
...	
0x1300	0x900
...	
0x1500	0x1000

[2 pts] Immediate addressing mode

Value loaded: 700 (direct value specified in the instruction)

[2 pts] Direct addressing mode

Value loaded: 0x800 (content at address 0x700)

[2 pts] Indirect addressing mode

Value loaded: 0x200 (content at address 0x800, which is the content of address 0x700)

5. [5 pts] A computer has a memory unit with 32 bits per word and has 16 general-purpose registers. The instruction set consists of 251 different operations. Some instructions have an opcode part, an address part, and a register part. Other instructions have an opcode part, and two address parts. Each instruction is stored in one word of memory.
   Explain how to determine the opcode number of bits and explain how many bits are needed to specify a memory address.

   Opcode and Register Bits:
   Number of operations: 251
   Opcode bits needed: $\lceil \log_2(251) \rceil = 8$ bits
   Number of registers:16
   Register bits needed: $\lceil \log_2(16) \rceil = 4$ bits

   Memory Address Bits:
   Instruction word: 32 bits.
   Case 1 (opcode + address + register): $8 + x + 4$ (for 16 registers) $\leqslant 32 \rightarrow x = 20$ bits.
   Case 2 (opcode + 2 addresses): $8 + 2x \leqslant 32 \rightarrow x = 12$ bits.
   Constraint: Choose the smaller value (12 bits) to fit both cases.

6. [2 pts] A non-pipelined system takes 200 ns to process a task. The same task can be processed in a 4-segment pipeline with a clock cycle of 10 ns. Determine the speedup ratio of the pipeline for 200 tasks. Give your answer to two decimal places of accuracy to the right of the decimal point.

   Non-pipelined: Total time = 200 tasks × 200 ns = 40,000 ns
   Pipelined:
   Time = (k + n - 1) × cycle time
   k = 4, n = 200, cycle = 10 ns
   Time = (4 + 200 - 1) × 10 = 2030 ns

   Speedup = 40000 / 2030 ≈ 19.70

7. [10 pts] Assume a computer that has 32-bit integers. Show how each of the following values would be stored sequentially in memory, starting at address 0x100, assuming each address holds one byte. Be sure to extend each value to the appropriate number of bits. You will need to add more rows (addresses) to store all five values.

0xB123456C

0x2BF876(padded to 32 bits as 0x002BF876)

0x8B0(padded to 32 bits as 0x000008B0)

0x1101(padded to 32 bits as 0x00001101)

0xFEDC123(padded to 32 bits as 0x0FEDC123)

Value (32-bit)	Address	Big Endian	Little Endian
0xB123456C	0x100	0xB1	0x6C
	0x101	0x23	0x45
	0x102	0x45	0x23
	0x103	0x6C	0xB1
0x002BF876	0x104	0x00	0x76
	0x105	0x2B	0xF8
	0x106	0xF8	0x2B
	0x107	0x76	0x00
0x000008B0	0x108	0x00	0xB0
	0x109	0x00	0x08
	0x10A	0x08	0x00
	0x10B	0xB0	0x00
0x00001101	0x10C	0x00	0x01
	0x10D	0x00	0x11
	0x10E	0x11	0x00
	0x10F	0x01	0x00
0x0FEDC123	0x110	0x0F	0x23
	0x111	0xED	0xC1
	0x112	0xC1	0xED
	0x113	0x23	0x0F